Mufeng Chen

Ph.D. Student and Graduate Research Assistant, Electrical and Computer Engineering

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Summary

- Research on memory systems and architecture for AI, including compute-in-memory and hybrid memory systems.

Education

Ph.D. Student — **Purdue University**, Electrical and Computer Engineering — West Lafayette, IN, USA — 2024 – present
- Advised by Prof. Haitong Li in the NanoX Lab.
- Research interests include AI memory systems, compute-in-memory, and hybrid memory systems.

Graduate Study — **Zhejiang University**, Electronic Engineering — Hangzhou, China — 2021 – 2024

B.Eng. — **Huazhong University of Science & Technology**, Optical and Electronic Information Engineering — Wuhan, China — 2017 – 2021

Experience

Purdue University, Graduate Research Assistant, NanoX Lab — West Lafayette, IN, USA
Advised by Prof. Haitong Li. — 2024 – present — 2 years

Zhejiang University, Research Assistant, RFNE Research Center — Hangzhou, China
Advised by Prof. Er-Ping Li. — 2021 – 2024 — 3 years

Research Internships

Purdue University, Research Intern, NanoX Lab — West Lafayette, IN, USA
Summer internship, advised by Prof. Haitong Li. — 2023 – 2023
- Research interests included compute-in-memory and hyperdimensional computing.

Rice University, Research Intern, SIMS Lab — Houston, TX, USA
Summer internship, advised by Prof. Kaiyuan Yang. — 2022 – 2022

Earlier Experience

Wuhan National Laboratory for Optoelectronics, Research Assistant — Wuhan, China
Advised by Prof. Chen Lin. — 2019 – 2019

Selected Publications

- M. Chen, L. Zheng, J.-Y. Lin, P. D. Ye, and H. Li. “Analog Multilevel eDRAM-RRAM CIM for Zeroth-Order Fine-tuning of LLMs.” 2025 IEEE International Memory Workshop, pp. 1–4. DOI: 10.1109/IMW61990.2025.11026966.
- L. Zheng, A. M. Bavani, M. Chen, et al. “PROTEUS: A 40 nm Programmable General-Purpose Digital Compute-In-Memory Accelerator With eNVM and Hierarchical ISA for Versatile Edge AI.” IEEE Journal of Solid-State Circuits, 2026. DOI: 10.1109/JSSC.2026.3694285.
- L. Zheng, Z. Wang, S. Du, M. Chen, A. M. Bavani, and H. Li. “Heterogeneous Compute-in-Memory Fabrics for Efficient, Scalable Edge Inference and Learning.” 2026 International Symposium on Quality Electronic Design. DOI: 10.1109/ISQED69900.2026.11534674.
- L. Zheng, A. M. Bavani, S. Du, T.-Y. Hsin, M. Chen, W.-S. Khwa, A. Lele, et al. “CENTAUR: A 38.5-TFLOPS/W 600MHz Floating-Point Digital Compute-In-Memory Engine with 40nm Fusion RRAM-eDRAM Macros Featuring 3D-MAC Operation.” 2025 IEEE Asian Solid-State Circuits Conference. DOI: 10.1109/A-SSCC67472.2025.11349414.
- E.-P. Li, H. Ma, M. Ahmed, T. Tao, Z. Gu, M. Chen, Q. Chen, D. Li, and W. Chen. “An Electromagnetic Perspective of Artificial Intelligence Neuromorphic Chips.” Electromagnetic Science, 1(3), 2023. DOI: 10.23919/emsci.2023.0015.
- C. Yu, Y. Du, M. Chen, A. Wang, G. Wang, and E. Li. “MAP-SNN: Mapping Spike Activities with Multiplicity, Adaptability, and Plasticity into Bio-Plausible Spiking Neural Networks.” Frontiers in Neuroscience, 16, 945037, 2022. DOI: 10.3389/fnins.2022.945037.

Technical Skills

Circuit & EDA: Cadence Virtuoso, Synopsys VCS, Design Compiler, Innovus

Programming: Python (PyTorch), C/C++, MATLAB, Verilog/SystemVerilog

Research areas: AI memory systems and architecture, compute-in-memory, hybrid memory systems, High-Bandwidth Flash research, hyperdimensional computing